A

PROJECT REPORT

ON

FOOD TRENDS - UNDERSTANDING CUSTOMER PREFERNCES IN F&B

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ABSTRACT

The Food and Beverage (F&B) industry generates large volumes of customer data through online platforms, which can be leveraged to understand customer preferences and improve business performance. This project focuses on analyzing restaurant data using Power BI to extract meaningful insights related to food trends, customer sentiment, product performance, and customer segmentation.

The dataset used in this project was sourced from Kaggle and includes information such as restaurant names, cuisines, ratings, votes, price range, and location. The data was cleaned and transformed using Power Query, and various measures were created using DAX to support analysis.

Multiple interactive dashboards were developed, including Food Trends Analysis, Sentiment Analysis, Product Insight Module, Customer Segmentation, and Strategy & Innovation Dashboard. These dashboards provide insights into popular cuisines, customer satisfaction levels, pricing patterns, and regional performance.

The analysis reveals that certain cuisines dominate customer engagement, most restaurants fall within mid-range pricing, and customer satisfaction is generally moderate. Additionally, customer segmentation highlights the presence of different user groups such as Loyal Fans, Critics, and Silent Users.

Overall, the project demonstrates how data analytics and visualization techniques can transform raw data into actionable insights that support decision-making in the F&B industry. The findings can help businesses improve customer experience, optimize product offerings, and identify opportunities for growth and innovation.

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1. INTRODUCTION

The Food and Beverage (F&B) industry is one of the fastest-growing sectors globally. Restaurants and food service businesses constantly compete to attract customers and maintain high satisfaction levels. With the growth of digital platforms such as Zomato and other food delivery services, large volumes of customer data are generated every day. This data provides valuable insights into customer preferences, restaurant performance, and emerging market trends.

Data analytics plays a crucial role in helping businesses understand customer behavior and improve decision-making. By analyzing customer ratings, engagement levels, and product preferences, businesses can identify opportunities for improvement and develop strategies to increase customer satisfaction and profitability.

This project focuses on analyzing restaurant data from the Zomato dataset to understand food trends, customer sentiment, segmentation patterns, and strategic opportunities in the F&B industry. Using Power BI as the primary business intelligence tool, the project transforms raw data into interactive dashboards that provide meaningful insights.

The dashboards developed in this project include Food Trends Analysis, Sentiment Analysis, Customer Segmentation, and Strategy & Innovation Insights. Each dashboard highlights a different aspect of customer behavior and restaurant performance. The analysis helps identify popular cuisines, customer satisfaction patterns, regional demand, and strategic opportunities for innovation.

Through this project, data analytics techniques such as data cleaning, transformation, visualization, and business intelligence reporting were applied to extract insights from the dataset. These insights can support restaurant owners, marketers, and business analysts in making data-driven decisions.

2. Project Overview

2.1 Project Background

With the increasing use of digital platforms for food ordering and restaurant discovery, customers frequently share their experiences through ratings and feedback. Platforms such as Zomato collect extensive data related to restaurant performance, customer preferences, and engagement levels.

This data provides an opportunity for businesses to analyze customer behavior and understand factors influencing their choices. Key aspects such as cuisine type, price range, location, and service quality play a significant role in determining customer satisfaction.

However, raw data alone does not provide direct insights. It needs to be processed, cleaned, and analyzed using appropriate tools and techniques. Business intelligence tools like Power BI enable analysts to visualize data, identify patterns, and generate actionable insights.

In this project, the Zomato dataset was used to explore customer behavior in the F&B industry. The project involves analyzing restaurant data to identify popular cuisines, evaluate customer satisfaction, and segment customers based on their engagement and feedback patterns.

The development of interactive dashboards allows users to explore data dynamically and gain insights into product performance, pricing strategies, and regional trends.

2.2 Project Objectives

The primary objective of this project is to analyze restaurant data and derive insights that can help businesses in the F&B industry improve their services, understand customer behavior, and identify market opportunities.

The specific objectives of the project are:

• To analyze food trends and identify popular cuisines based on customer engagement.
•To evaluate customer sentiment using rating-based sentiment classification.
• To segment customers into behavioral categories such as Loyal Fans, Critics, and Silent Users.
• To analyze restaurant performance across different cities and price ranges.
• To identify strategic opportunities for innovation and marketing using data insights.
• To build interactive dashboards that allow dynamic exploration of restaurant data.

By achieving these objectives, the project aims to demonstrate how business intelligence tools can convert raw data into actionable insights that support strategic decision-making.

2.3 Project Description

This project focuses on analyzing restaurant data using Power BI to extract meaningful insights related to customer preferences and restaurant performance.

The project consists of multiple analytical modules that together provide a comprehensive understanding of the F&B industry.

The first module focuses on identifying food trends by analyzing which cuisines attract the most customer engagement. This helps businesses understand popular food categories and market demand.

The second module involves sentiment analysis. Customer ratings are used to determine whether customers have a positive, neutral, or negative perception of restaurants. This analysis helps businesses identify areas where service improvements may be needed.

The third module focuses on customer segmentation. Customers are categorized into different behavioral groups based on engagement and satisfaction levels. This allows businesses to understand their customer base and design targeted marketing strategies.

The final module focuses on strategy and innovation insights. This dashboard highlights trending cuisines, regional demand patterns, and customer satisfaction trends to help businesses identify opportunities for growth and product innovation.

3. Dataset Overview and Data Collection

3.1 Dataset Overview

The dataset used in this project was obtained from Kaggle and is based on restaurant data collected from the Zomato platform. The dataset contains structured information about restaurants, customer engagement, and customer feedback, which makes it suitable for analyzing customer preferences in the Food and Beverage (F&B) industry.
The dataset includes multiple attributes that provide insights into restaurant performance and customer behavior. Key attributes used in this project are:

- Restaurant Name – identifies individual restaurants
- City – represents the location of restaurants
- Cuisines – indicates the type of food offered
- Votes – represents customer engagement
- Aggregate Rating – indicates customer satisfaction
- Price Range – categorizes restaurants based on cost
- Average Cost for Two – estimates customer spending
- Online Delivery – indicates availability of delivery services

The dataset enables analysis of various aspects such as food trends, pricing patterns, customer satisfaction, and regional demand.

3.2 Data Collection

The data was collected from Kaggle, which provides publicly available datasets for data analysis and research. The Zomato dataset used in this project is derived from real-world restaurant data collected through online platforms.

The dataset represents customer interactions in the form of ratings and votes, which reflect customer preferences and satisfaction levels. This data is valuable for understanding how customers interact with restaurants and which factors influence their choices.

The data was downloaded in CSV format and imported into Power BI for further processing and analysis.

4. Data Extraction Process

4.1 Data Ingestion in Power BI

The dataset was imported into Power BI using the "Get Data" feature. The CSV file was loaded into Power Query Editor, where initial data inspection was performed.

Steps involved:

- Importing dataset using "Get Data → CSV"
- Loading data into Power Query
- Previewing dataset structure
- Identifying relevant columns

4.2 Data Selection and Filtering

Not all columns in the dataset were relevant to the analysis. Therefore, only important attributes were selected, and unnecessary columns were removed.

Selected columns include:

- Restaurant Name
- City
- Cuisines
- Votes
- Aggregate Rating
- Price Range
- Average Cost for Two

This step helps in improving performance and simplifying the dataset.

4.3 Initial Data Inspection

Before cleaning, the dataset was examined to identify:

- Missing values
- Incorrect data types
- Duplicate records
- Inconsistent entries

This inspection helped in planning the data cleaning process.

5. Data Cleaning and Data Preparation

5.1 Data Cleaning and Transformation

Data cleaning was performed using Power Query Editor in Power BI. This step ensures that the dataset is accurate, consistent, and ready for analysis.

5.2 Handling Inconsistent Data

Inconsistent data can lead to incorrect analysis. In this dataset:

- Some price range values were missing or invalid
- Cuisine names had multiple combinations
- Some rating or vote entries were zero or incorrect

These inconsistencies were handled by:

- Removing invalid rows
- Filtering out zero or unrealistic values
- Standardizing data where possible

5.3 Data Type Enforcement

Correct data types are essential for proper calculations and visualization.

The following conversions were applied:

- Votes → Whole Number
- Aggregate Rating → Decimal Number
- Price Range → Whole Number
- Cuisines → Text

Ensuring correct data types helps avoid calculation errors.

5.4 Removing Duplicates and Irrelevant Columns

Duplicate records can distort analysis results. Therefore:

- Duplicate rows were checked and removed
- Unnecessary columns such as IDs and URLs were removed

This step reduced dataset size and improved clarity.

5.5 Importance of Data Preparation

Data preparation is one of the most important steps in data analysis.

Benefits include:

- Improves data accuracy
- Ensures reliable results
- Reduces errors in calculations
- Enhances visualization quality

Clean and structured data is essential for building effective dashboards.

6. Data Transformation and Measure Creation

After data cleaning, the next step is data transformation, where the dataset is converted into a structured format suitable for analysis. In this project, data transformation was performed using Power BI and DAX to create meaningful metrics and classifications.

This step helps in converting raw data into useful information that supports business analysis and visualization.

6.1 Data Transformation Techniques

Several transformation techniques were applied:

- Renaming columns for better readability
- Filtering invalid or missing values
- Standardizing data formats
- Creating new derived columns

Two important derived columns were created:

- **Sentiment Column** → Categorizes ratings into Positive, Neutral, and Negative
- **Customer Segment Column** → Classifies users into Loyal Fans, Critics, and Silent Users

These transformations help in better understanding customer behavior.

6.2 Measure Creation Using DAX

DAX was used to create key performance measures:

- **Total Sales** = SUM(Votes) → Represents customer engagement
- **Average Rating** = AVERAGE(Aggregate Rating) → Indicates satisfaction level
- **Total Revenue** = Average Cost × Votes → Estimates performance
- **Segment Counts** → Number of Loyal Fans, Critics, and Silent Users

These measures are used as KPIs in dashboards.

6.3 Importance of Data Transformation

Data transformation is important because it:

- Improves data usability
- Enables meaningful analysis
- Supports dashboard creation
- Helps in better decision-making

Without transformation, raw data cannot provide actionable insights.

7. Data Analysis and Visualization

7.1 Dashboard Design

The dashboard was designed with the objective of presenting complex data in a simple, clear, and interactive manner. The design follows key principles of business intelligence such as clarity, consistency, and usability.

The layout is structured in a top-to-bottom approach. Key performance indicators (KPIs) are displayed at the top to provide a quick summary of overall performance. Detailed visualizations are placed below to allow deeper analysis of customer preferences and restaurant trends.

Consistent color schemes and formatting were used to ensure readability and maintain visual appeal. Interactive filters such as city and cuisine were included to enable dynamic data exploration.

The dashboard focuses on answering important business questions such as:

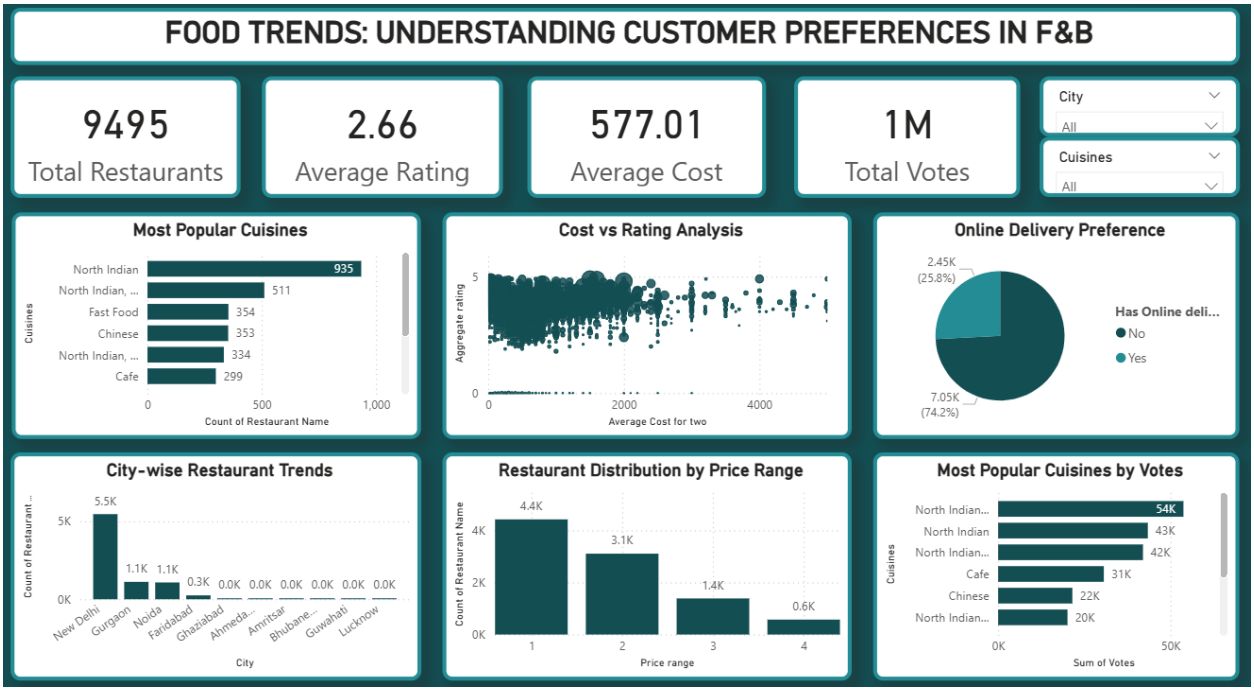
- Which cuisines are most popular?
- How does pricing affect customer satisfaction?
- Which cities generate the highest demand?
- What are the customer behavior patterns?

Overall, the design ensures that users can quickly understand insights and make informed decisions.

7.2 Dashboard Overview

The Food Trends Dashboard provides a comprehensive overview of customer preferences, restaurant performance, and market trends in the F&B industry.

It combines multiple visualizations such as KPI cards, bar charts, scatter plots, and pie charts to present data in an intuitive format.



7.3 Explanation of Visuals

➢ KPI Cards

The dashboard includes four key performance indicators:

- **Total Restaurants (9495)**
  Represents the total number of restaurants in the dataset.
- **Average Rating (2.66)**
  Indicates the overall customer satisfaction level.
- **Average Cost (577.01)**
  Shows the average spending for two people.
- **Total Votes (1M)**
  Represents total customer engagement.

These KPIs provide a quick summary of overall market performance.

➢ Most Popular Cuisines

This bar chart shows the number of restaurants for each cuisine type.

Insight:
North Indian cuisine has the highest presence, followed by fast food and Chinese cuisine.

This indicates strong demand for traditional and commonly available cuisines.

- **Cost vs Rating Analysis**

This scatter plot shows the relationship between average cost and customer ratings.

Insight:
Most restaurants are clustered in mid-cost and mid-rating ranges, indicating that higher prices do not always guarantee better ratings.

This helps businesses understand pricing and customer satisfaction relationships.

- **Online Delivery Preference**

This pie chart shows the distribution of restaurants offering online delivery.

Insight:
A large percentage of restaurants provide delivery services, indicating the importance of convenience in customer decision-making.

- **City-wise Restaurant Trends**

This bar chart shows the distribution of restaurants across cities.

Insight:
New Delhi has the highest number of restaurants, followed by other major cities.

This highlights regional concentration of restaurant businesses.

- **Restaurant Distribution by Price Range**

This chart shows how restaurants are distributed across different price ranges.

Insight:
Most restaurants fall in price range 1 and 2, indicating a preference for budget and mid-range dining.

➢ **Most Popular Cuisines by Votes**

This chart shows customer engagement across cuisines based on votes.

Insight:
North Indian cuisine receives the highest engagement, indicating strong customer interest.

7.4 Interactive Features

The dashboard includes filters for:

- City
- Cuisines

These filters allow users to explore data dynamically and analyze specific segments based on location and food preferences.

7.5 Key Insights from the Dashboard

- North Indian cuisine dominates both availability and customer engagement
- Most restaurants fall within budget and mid-range pricing
- Customer ratings are generally moderate (around 2.5–3)
- Online delivery is widely adopted
- New Delhi is a major hub for restaurant activity

These insights help businesses understand customer preferences and market trends.

7.6 Importance of Data Visualization in Business Decision-Making

Data visualization plays a crucial role in transforming raw data into meaningful insights. It allows businesses to interpret complex datasets quickly and effectively.

The importance of visualization includes:

- Simplifies complex data for easy understanding
- Helps identify trends and patterns
- Supports faster decision-making
- Improves communication of insights
- Enables interactive data exploration

In this project, visualization techniques helped convert restaurant data into actionable insights that support strategic planning in the F&B industry.

8. Sentiment Analysis

8.1 Introduction

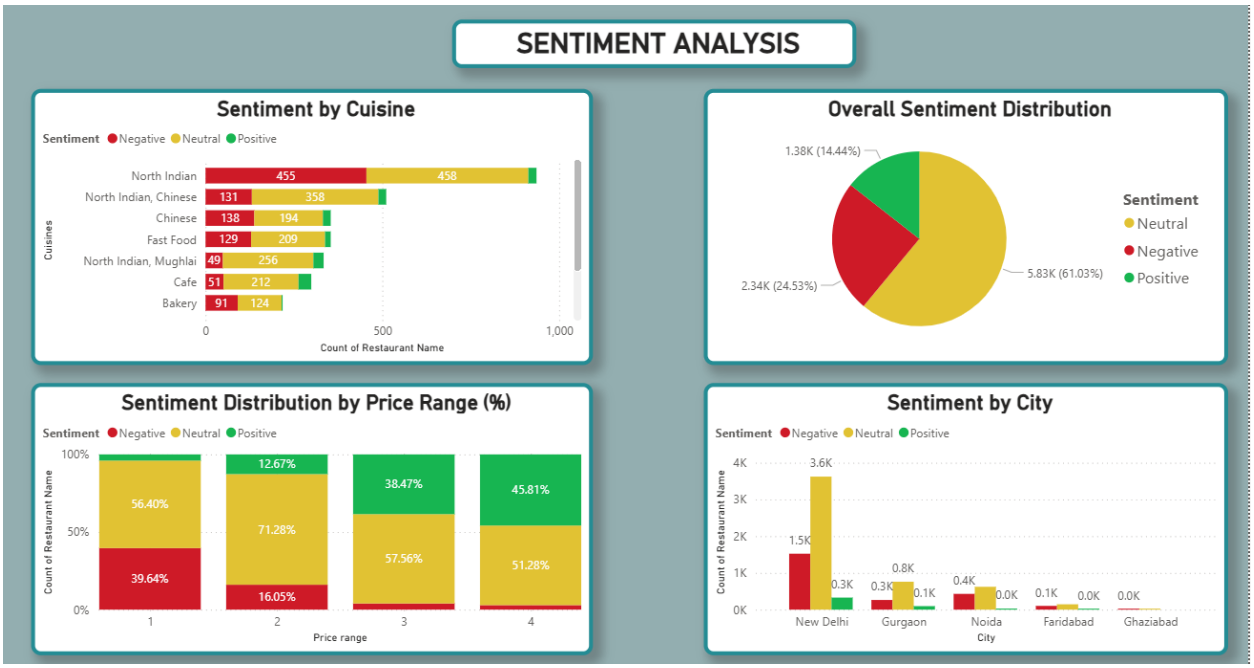
Sentiment analysis is used to understand customer perception and satisfaction based on their feedback. In the F&B industry, customer sentiment plays a crucial role in evaluating service quality, food taste, and overall dining experience.

In this project, sentiment analysis was performed using customer ratings. Since textual reviews were not available, sentiment was derived from numerical ratings and classified into three categories:

- **Positive Sentiment** → Rating ≥ 4
- **Neutral Sentiment** → Rating between 2.5 and 3.9
- **Negative Sentiment** → Rating < 2.5

This classification helps convert numerical data into meaningful insights about customer satisfaction.

8.2 Dashboard Overview



The Sentiment Analysis Dashboard provides a comprehensive view of customer sentiment across cuisines, price ranges, and cities. It helps identify patterns in customer satisfaction and highlights areas that require improvement.

8.3 Explanation of Visuals

- **Sentiment by Cuisine**

This stacked bar chart shows the distribution of sentiment across different cuisines.

Insight:
North Indian cuisine has the highest number of reviews, with a large portion falling under neutral and positive sentiment. Some cuisines show noticeable negative sentiment, indicating areas where quality or service may need improvement.

- **Overall Sentiment Distribution**

This pie chart shows the overall distribution of customer sentiment.

Insight:
Most restaurants fall under **neutral sentiment**, followed by positive sentiment, while a smaller percentage falls under negative sentiment. This indicates that customer satisfaction is moderate, with scope for improvement.

- **Sentiment Distribution by Price Range (%)**

This 100% stacked column chart shows how sentiment varies across different price ranges.

Insight:
Lower price ranges have higher neutral and negative sentiment, while higher price ranges show a greater proportion of positive sentiment. This suggests that customers may associate higher pricing with better quality and experience.

➢ **Sentiment by City**

This bar chart shows customer sentiment across different cities.

Insight:
Cities like New Delhi show higher engagement with a mix of sentiment categories. Some cities have relatively lower engagement but show consistent sentiment patterns.

8.4 Key Insights from Sentiment Analysis

- Majority of customer feedback falls under **neutral sentiment**
- Higher price ranges tend to have **better customer satisfaction**
- Certain cuisines show **high demand but mixed sentiment**
- Major cities generate higher engagement and varied sentiment

These insights help businesses identify areas where service quality can be improved.

8.5 Business Importance of Sentiment Analysis

Sentiment analysis is essential for understanding customer experience and improving business performance.

Its importance includes:

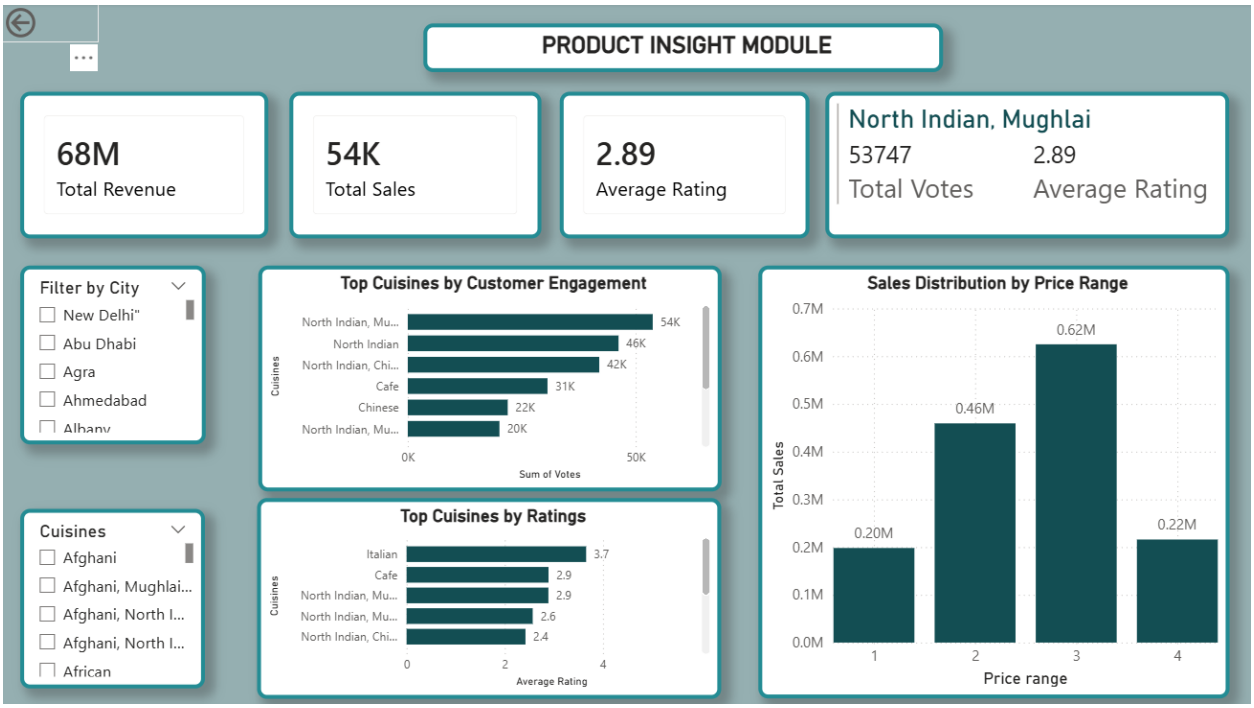
- Identifying customer satisfaction levels
- Detecting areas of improvement in service and product quality
- Understanding customer perception across different price segments
- Supporting decision-making for product improvement
- Enhancing customer retention strategies

In this project, sentiment analysis helps transform customer ratings into actionable insights that support better decision-making in the F&B industry.

The results of this analysis help restaurants improve service quality, optimize pricing strategies, and enhance customer experience.

9. Product Insight Module

9.1 Dashboard Overview



The Product Insight Module focuses on analyzing product performance in the F&B industry using key metrics such as revenue, sales, ratings, and customer engagement. This dashboard helps identify high-performing cuisines, understand pricing impact, and evaluate customer preferences.
It provides a quick overview of business performance through KPIs and detailed visualizations for deeper analysis.

9.2 Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- **Total Revenue (68M)**

Represents estimated revenue based on customer engagement and pricing.

- **Total Sales (54K)**

Indicates total customer interactions (votes), representing demand.

- **Average Rating (2.89)**

Shows overall customer satisfaction level.

- **Top Cuisine (North Indian, Mughlai)**

Displays the highest performing cuisine along with its total votes and rating.
These KPIs provide a quick understanding of product performance.

9.3 Explanation of Visuals

➢ **Top Cuisines by Customer Engagement**

This bar chart shows cuisines based on total votes.

Insight:

North Indian and Mughlai cuisines generate the highest engagement, indicating strong customer demand for these categories.

➢ **Top Cuisines by Ratings**

This chart ranks cuisines based on average ratings.

Insight:

Some cuisines (like Italian) have higher ratings but may not have the highest engagement. This shows that popularity and satisfaction are not always the same.

➢ **Sales Distribution by Price Range**

This chart shows how total sales are distributed across price ranges.

Insight:

Price range 2 and 3 generate the highest sales, indicating that customers prefer mid-range pricing over very low or very high prices.

9.4 Filters and Interactivity

The dashboard includes filters for:

- City
- Cuisines

These filters allow users to analyze product performance across different regions and categories dynamically.

9.5 Key Insights from Product Analysis

- Mid-range pricing generates the highest sales
- North Indian cuisine dominates in customer engagement
- High-rated cuisines are not always the most popular
- Customer demand is concentrated in specific product categories

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9.6 Business Importance of Product Insights

Product insights are essential for:

- Identifying high-performing cuisines
- Optimizing pricing strategies
- Improving product quality based on ratings
- Understanding customer preferences
- Supporting menu and marketing decisions

10. Customer Segmentation

Customer segmentation is the process of dividing customers into distinct groups based on their behavior, preferences, and engagement patterns. In the Food and Beverage (F&B) industry, segmentation helps businesses understand different types of customers and design targeted strategies to improve satisfaction and retention.

In this project, customers were segmented based on their interaction with restaurants using metrics such as ratings and votes. This approach helps identify how different groups of customers behave and what influences their preferences.

The segmentation allows businesses to focus on specific customer groups and improve overall service quality.

10.1 Customer Segmentation Criteria

Customers were segmented based on their engagement (votes) and satisfaction (ratings). This classification helps in understanding different customer behaviors.

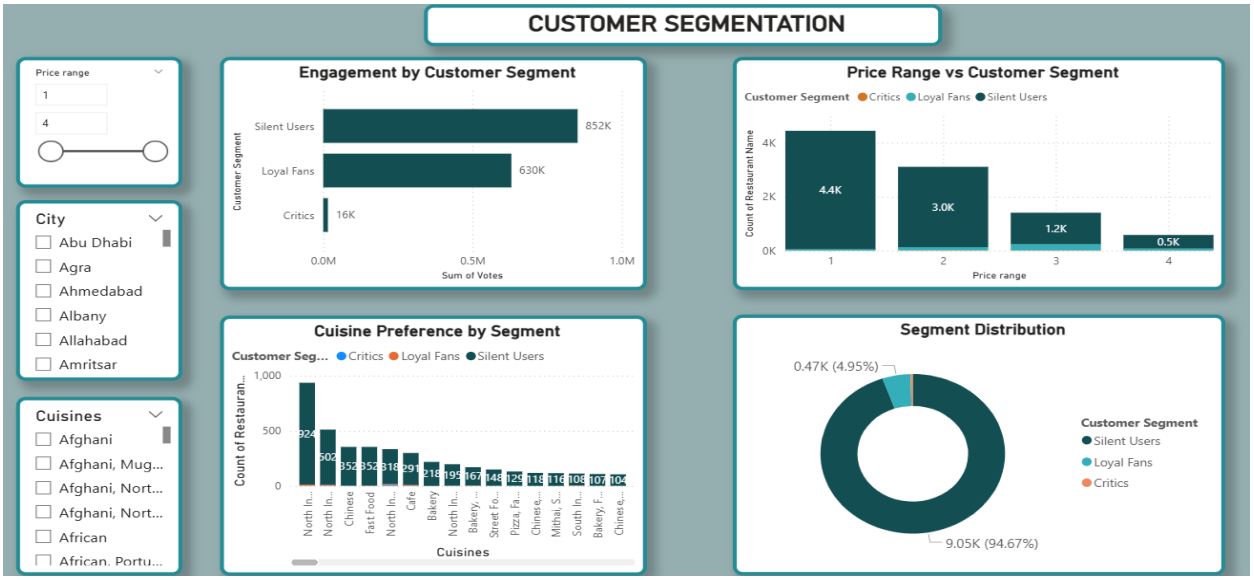
The segments are defined as:

- **Loyal Fans** → Customers with high ratings and strong engagement
- **Critics** → Customers who provide low ratings and express dissatisfaction
- **Silent Users** → Customers with high interaction but neutral or limited feedback

This segmentation was implemented using calculated columns in Power BI.

10.2 Dashboard Overview

The Customer Segmentation dashboard provides insights into customer behavior across different dimensions such as engagement, price range, and cuisine preferences. It also includes filters for city, cuisine, and price range to enable interactive analysis.



10.3 Explanation of Visuals

➢ Engagement by Customer Segment

This bar chart shows total engagement (votes) across customer segments.

Insight:
Silent Users have the highest engagement (around 852K), followed by Loyal Fans (630K), while Critics contribute very little.
This indicates that most users interact with restaurants but do not provide strong feedback.

➢ Price Range vs Customer Segment

This chart shows how customer segments are distributed across different price ranges.

Insight:
Most activity is concentrated in **price ranges 1 and 2**, indicating a strong preference for budget and mid-range restaurants.
Higher price ranges have fewer customers across all segments.

➢ Cuisine Preference by Segment

This chart shows how different cuisines attract various customer segments.

Insight:
North Indian cuisine dominates across all segments, especially Silent Users.
Popular cuisines attract a mix of all customer types, but engagement is highest from Silent Users.

- **Segment Distribution**

This donut chart shows the overall proportion of customer segments.

Insight:
Silent Users form the majority (around 94%), while Loyal Fans and Critics represent a small portion of the dataset.
This highlights a gap in active feedback from customers.

10.4 Key Insights from Customer Segmentation

- Silent Users dominate customer engagement
- Loyal Fans are fewer but valuable for business growth
- Critics are minimal but important for identifying issues
- Budget and mid-range restaurants attract most customers
- Popular cuisines attract all types of users

10.5 Business Importance of Customer Segmentation

Customer segmentation helps businesses in multiple ways:

- Enables targeted marketing strategies
- Helps convert Silent Users into Loyal Fans
- Identifies dissatisfied customers (Critics) for improvement
- Supports personalized customer experience
- Improves customer retention and engagement

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11. Strategy and Innovation

11.1 Dashboard Overview

The Strategy and Innovation dashboard focuses on identifying business opportunities, customer trends, and areas for improvement in the F&B industry. It combines sentiment, pricing, customer feedback, and regional performance to support strategic decision-making.

The dashboard helps answer key business questions such as:

- How does pricing impact customer sentiment?
- What is the overall distribution of customer ratings?
- Which cuisines are trending?
- Which regions generate the highest engagement?

11.2 Explanation of Visuals

➢ Sentiments vs Price

This stacked column chart shows the relationship between customer sentiment and price range.

Insight:
Lower price ranges (1 and 2) have higher customer engagement but show a mix of neutral and negative sentiment.
Higher price ranges (3 and 4) show relatively better positive sentiment.

This indicates that customers expect better quality and service as price increases.

- **Rating Distribution**

This histogram shows the distribution of aggregate ratings across restaurants.

Insight:
Most ratings are concentrated between **3 and 4**, indicating moderate customer satisfaction.
Very few restaurants achieve ratings close to 5.

This suggests that there is significant scope for improving customer experience.

- **Trending Cuisines**

This bar chart shows cuisines based on total customer engagement (votes).

Insight:
North Indian and Mughlai cuisines receive the highest engagement, followed by Chinese and Cafe categories.

This highlights strong demand for traditional and popular cuisines.

- **Regional Performance**

This treemap shows customer engagement across different cities.

Insight:
New Delhi dominates with the highest engagement, followed by cities like Gurgaon and Noida.
Other cities show comparatively lower activity.

This indicates that certain regions have higher market potential.

11.3 Key Insights from Strategy Analysis

- Customer satisfaction is mostly moderate, not extremely high
- Higher price ranges tend to have better sentiment
- Popular cuisines dominate customer engagement
- Certain cities contribute significantly more to overall activity
- There is a gap between demand and satisfaction in some segments

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11.4 Strategic Recommendations

Based on the analysis, the following strategies can be suggested:

- Improve service quality in lower price ranges to increase satisfaction
- Focus on expanding high-demand cuisines
- Target high-performing regions for business expansion
- Enhance customer experience to improve ratings above 4
- Convert neutral customers into positive customers through better service

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11.5 Role in Business Decision-Making

This dashboard plays a crucial role in supporting strategic decisions:

- Helps identify market trends and opportunities
- Supports product and menu optimization
- Guides pricing strategies
- Assists in regional expansion planning
- Improves customer satisfaction strategies

12. Conclusion

This project demonstrates how data analytics and business intelligence tools can transform raw restaurant data into meaningful insights. Through the use of Power BI dashboards, various aspects of customer behavior, restaurant performance, and market trends were analyzed.

The analysis revealed important insights regarding customer preferences, engagement patterns, and satisfaction levels. The Food Trends dashboard helped identify popular cuisines and regional demand patterns. Sentiment analysis provided insights into customer perception and satisfaction. Customer segmentation allowed the identification of different user personas, while the Strategy and Innovation dashboard highlighted potential growth opportunities.

The results of this project show that data-driven decision-making can help businesses in the F&B industry improve customer experience, optimize pricing strategies, and identify new market opportunities.

Future improvements could include integrating real-time data, applying machine learning models for predictive analysis, and incorporating textual customer reviews for more advanced sentiment analysis.

Overall, this project highlights the importance of data analytics in understanding customer behavior and supporting strategic business decisions.